

Supplementary information: Cell-free protein synthesis as a method to rapidly screen machine learning-directed protease variants

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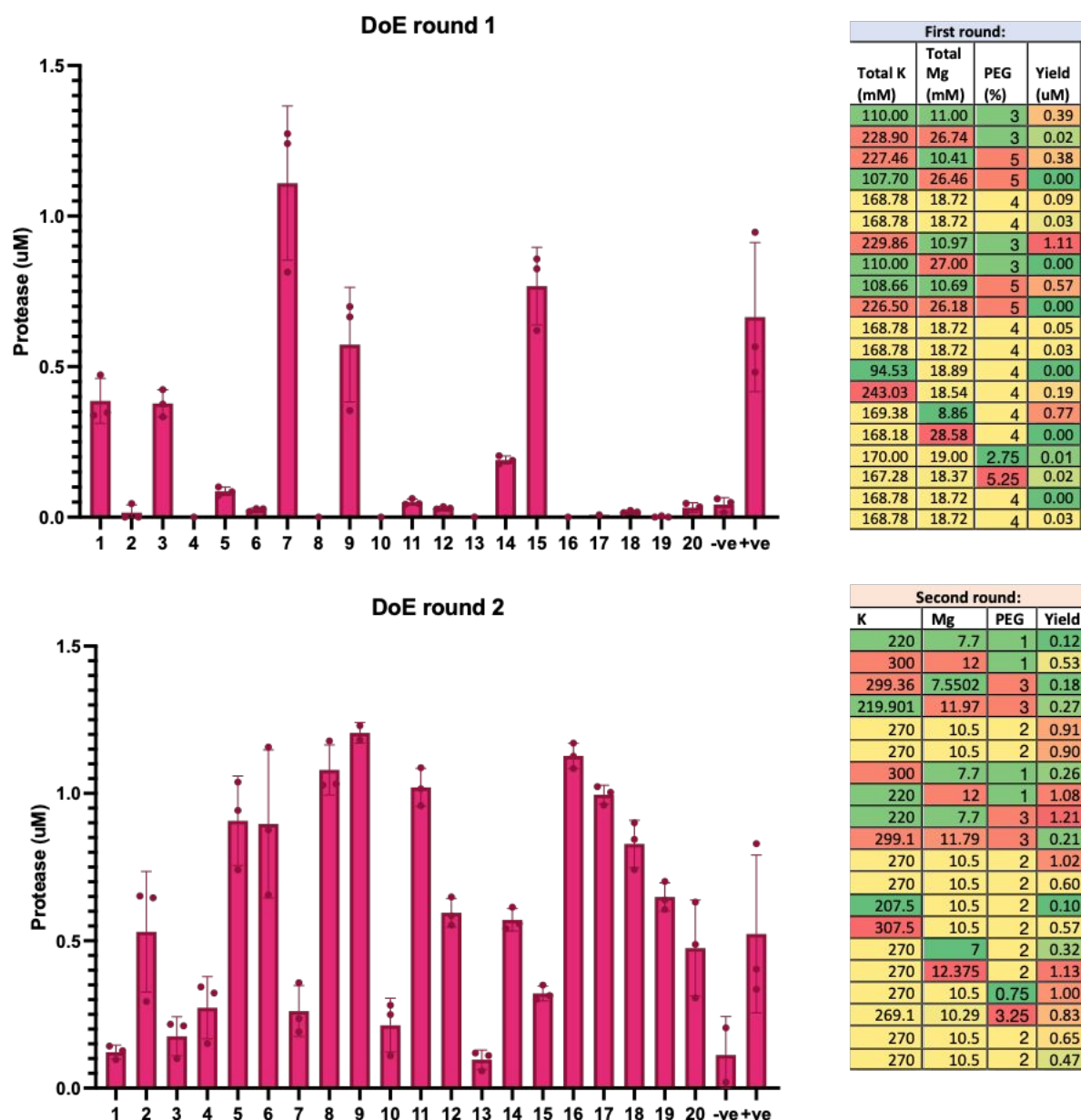


Figure S1. A Design of Experiments approach to optimise protease expression in CFPS. The experimental design of Central Composite Design was utilised, with axial points. Mg, K, and PEG were selected as variables to assess. The range of concentrations to be tested for each was decided by a literature review. The amount of protease made in each condition was assessed by an activity assay and compared to a purified calibration of protease against FRET substrate (S2). Round one identified conditions that were more favourable for protease expression (as seen by comparison with the +ve sample, with original energy solution components). To further optimise the conditions, another round of DoE was planned and completed, finding a condition that increased protease expression from ~0.5 μM to ~1 μM . These conditions were used for all subsequent CFPS experiments.

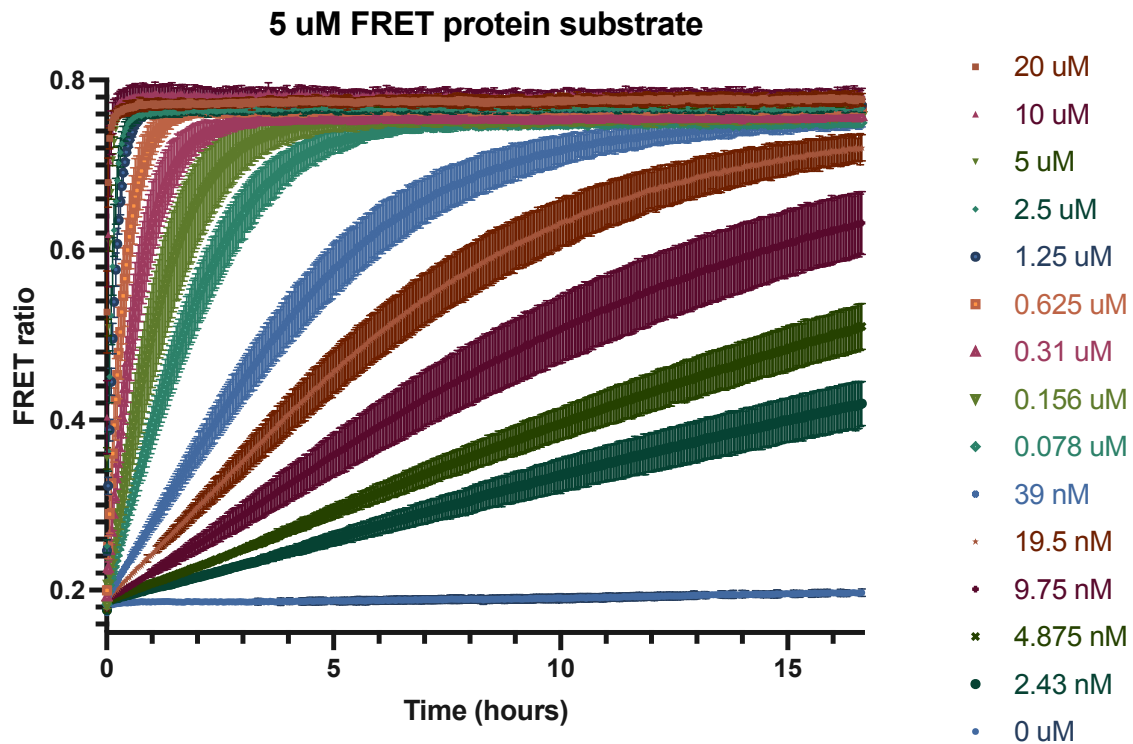
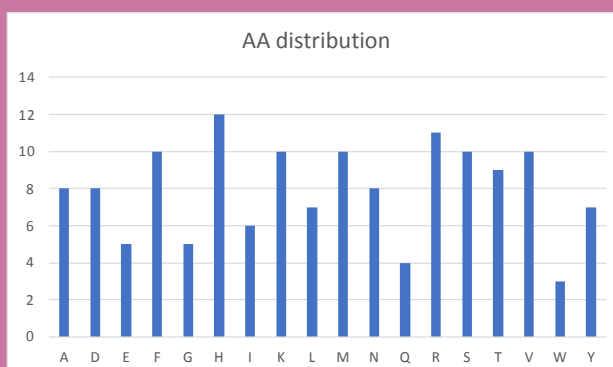
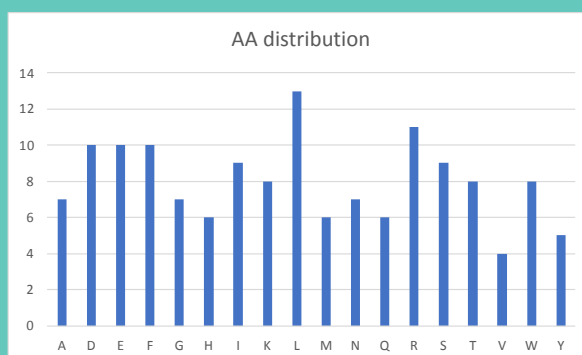


Figure S2. Calibration of FRET substrate against known concentrations of purified Con1 protease. FRET substrate was mixed together with protease in buffer, to give a final FRET concentration of 5 μ M. This was immediately measured in a plate reader for fluorescence and subsequent assessment of FRET over time.

Region A:

Mutant no:	Residue:		
	167	169	172
1	E	I	N
2	T	H	T
3	R	M	F
4	L	L	A
5	R	R	L
6	W	A	K
7	R	K	L
8	W	A	F
9	G	F	F
10	Q	L	F
11	H	T	V
12	G	Y	R
13	V	M	N
14	V	T	D
15	Y	F	N
16	W	L	G
17	Q	R	E
18	E	R	W
19	M	S	I
20	D	L	G
21	E	D	W
22	K	S	F
23	K	I	W
24	D	A	G
25	G	S	Y
26	L	T	K
27	R	A	H
28	H	M	E
29	Q	V	T
30	N	K	D
31	E	H	S
32	S	R	F
33	W	R	E
34	A	L	F
35	S	W	S
36	I	E	N
37	I	L	H
38	L	M	D
39	I	A	N
40	L	M	F
41	I	Y	I
42	S	T	E
43	D	D	T
44	E	Q	K
45	N	Q	Y
46	S	D	R
47	Q	L	G
48	K	D	I



Region B:

Mutant no:	Residue:		
	217	218	219
1	H	V	L
2	Y	L	I
3	E	G	K
4	N	N	R
5	Q	T	H
6	R	M	H
7	W	A	K
8	H	S	Y
9	H	M	Y
10	L	N	M
11	A	M	G
12	K	F	W
13	F	H	S
14	S	H	F
15	A	V	E
16	M	D	H
17	I	V	T
18	S	K	Q
19	D	H	R
20	R	S	K
21	I	E	D
22	A	R	V
23	W	R	H
24	A	R	F
25	M	D	Y
26	S	V	S
27	G	Y	Y
28	G	N	H
29	F	R	F
30	R	T	A
31	T	K	K
32	F	S	L
33	R	K	N
34	M	T	F
35	Y	G	D
36	K	M	I
37	A	K	T
38	F	L	T
39	Y	V	D
40	F	L	V
41	I	E	N
42	V	A	N
43	M	Q	H
44	I	S	D
45	E	V	T
46	N	Q	V
47	S	D	T
48	M	R	L

Figure S3. Identity of all random mutants. 48 random mutants were created for both region A and region B by a script which compiled a randomised listed of amino acids into a 3 x 48 table. At this stage of the design process, we removed proline and cysteine as possibilities for mutagenesis. The representation of all other amino acids was verified as seen in the bar plots.

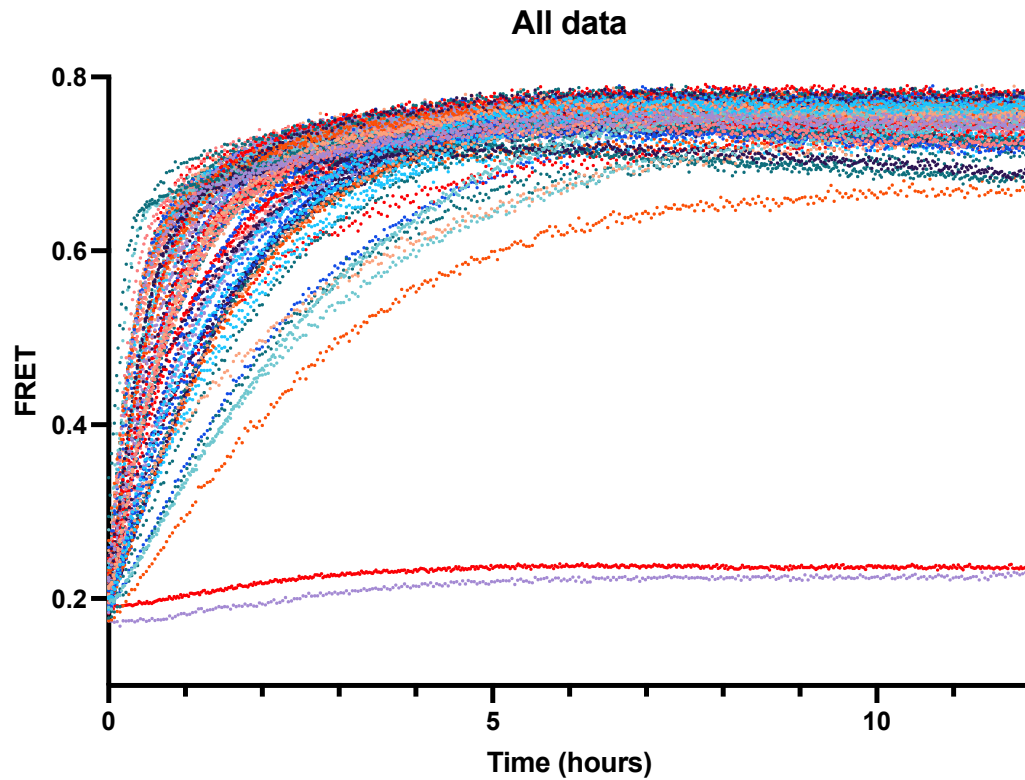


Figure S4. Example time-course data of FRET for 48 B mutants after production by CFPS.

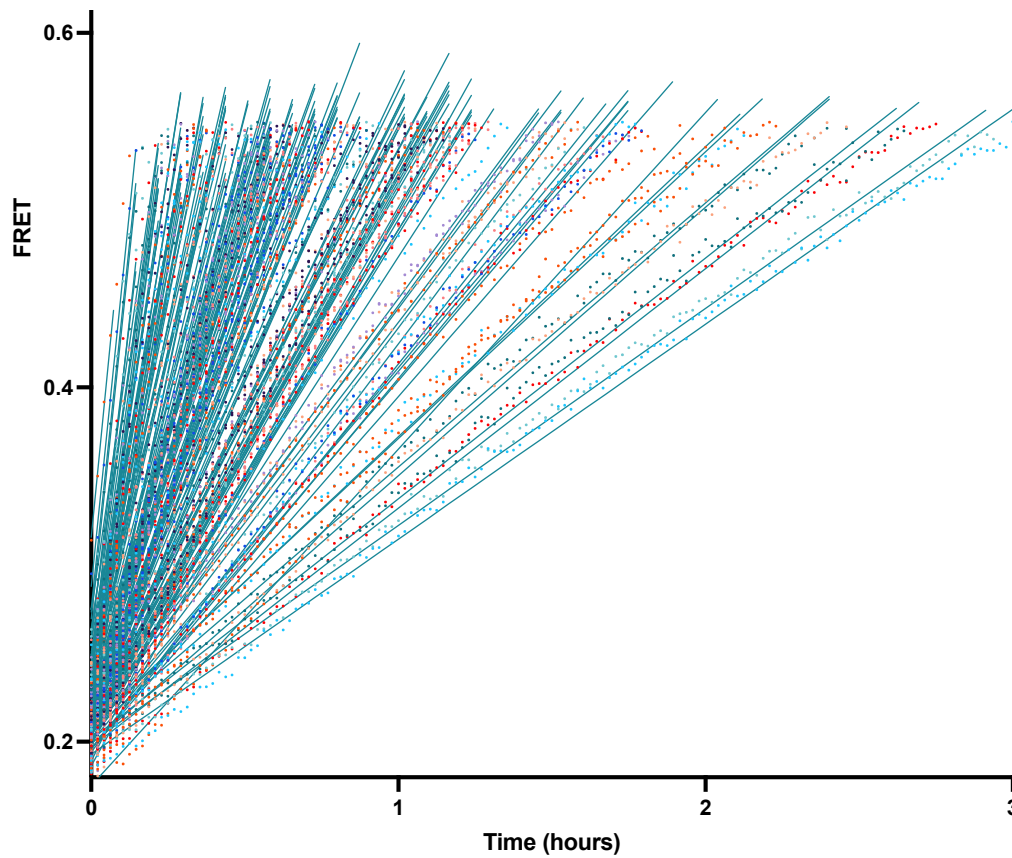


Figure S5. Data from S4 truncated at FRET = 0.55. Plot is cropped to Time = 3 hours to allow closer examination of data points and linear fit. We found this data processing step allowed for accurate scoring of mutant activity, as shown further in Figure S6.

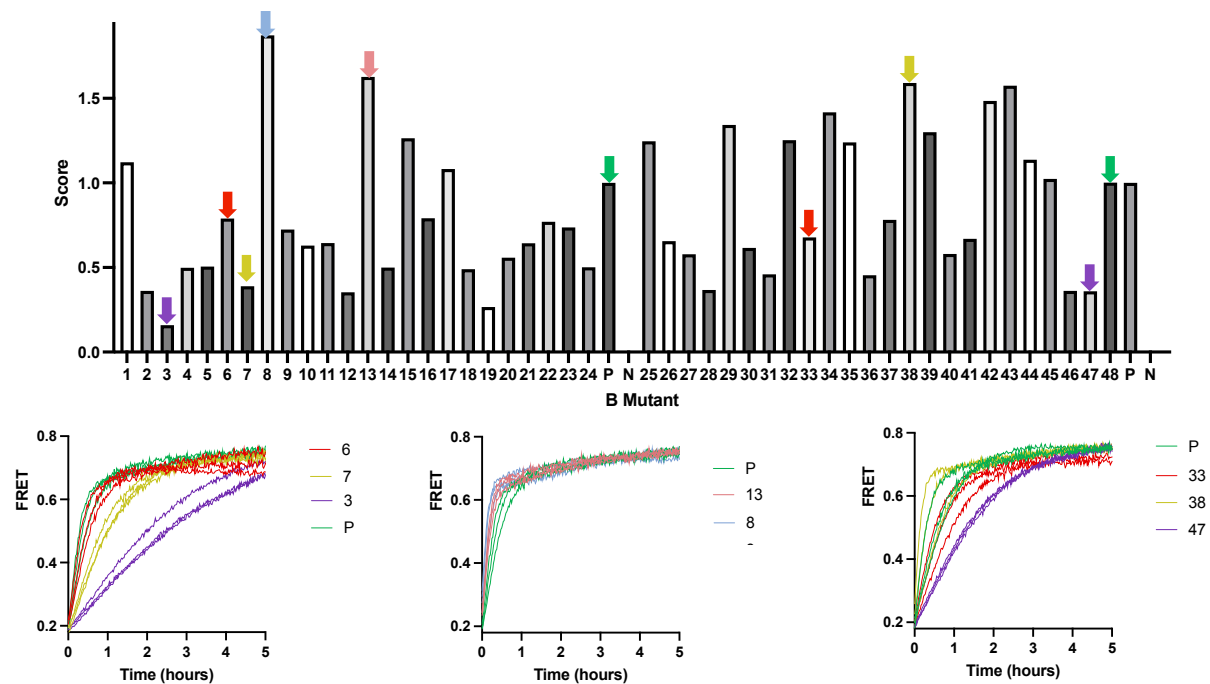


Figure S6. Assessment of final fitness score accuracy against original FRET time-course data. Final scores (average of 3 replicates) are shown in the bar chart for all region B mutants. Below, line charts show individual replicates for selected mutants, the colour of which is then matched to their identity and to the corresponding bar.

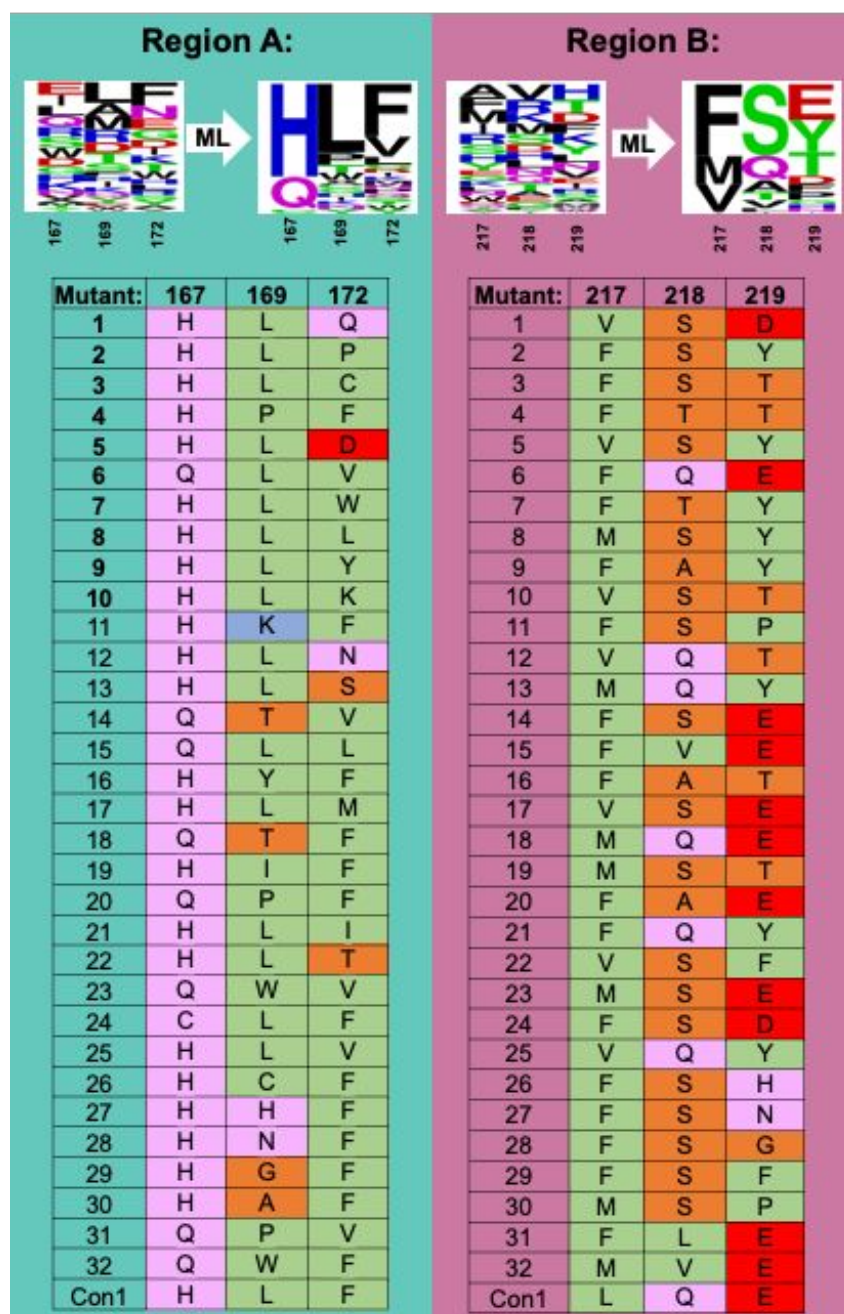


Figure S7. Identity of mutants suggested by ALDE workflow. Sequence logos of the amino acid identities of the random mutants, compared with machine learning suggested mutants are shown. For region A, only the first 10 of these suggested mutants were tested. For region B, all 32 were screened. The identity of each variant at the residues to be mutated is indicated in the table, with amino acids coloured based on type: green = hydrophobic, pink = polar, red = negatively charged, orange = small non-polar, blue = positively charged.

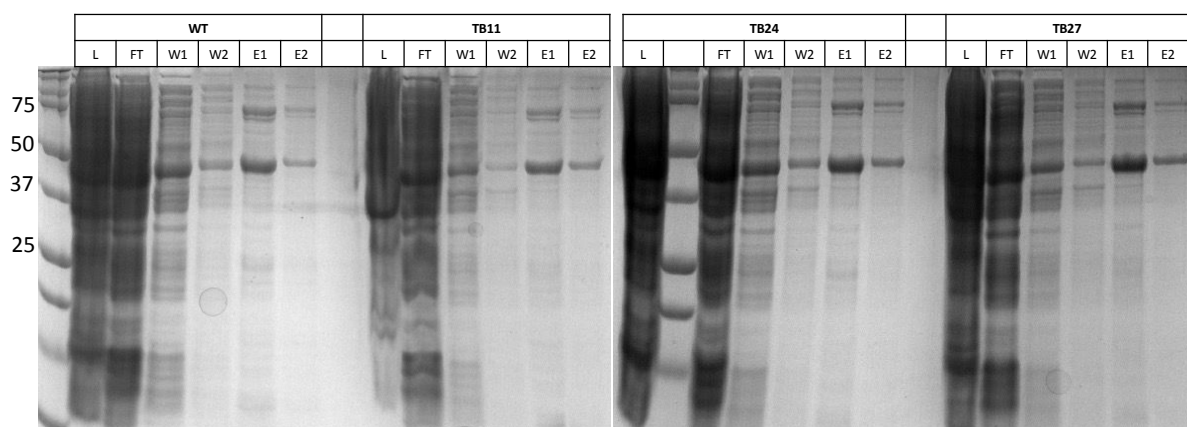


Figure S8. Affinity purification of four protease variants. Indicated by the lane title, different samples from the purification process were loaded: L (lysate), FT (flow-through), W1 (wash 1), W2 (wash 2), E1 (elution 1), E2 (elution 2). Expected MW = 42 kDa.

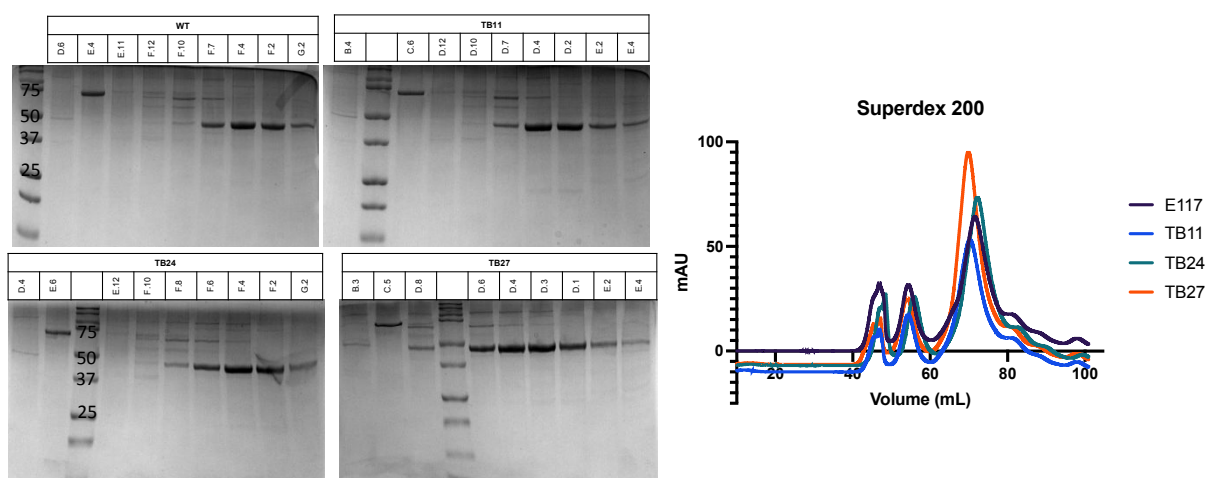


Figure S9. Size exclusion chromatography of four protease variants. Fractions eluted from the column were run on SDS-PAGE to assess purity. Absorbance from the SEC run is shown on the line plot. The largest peak contained the proteases.

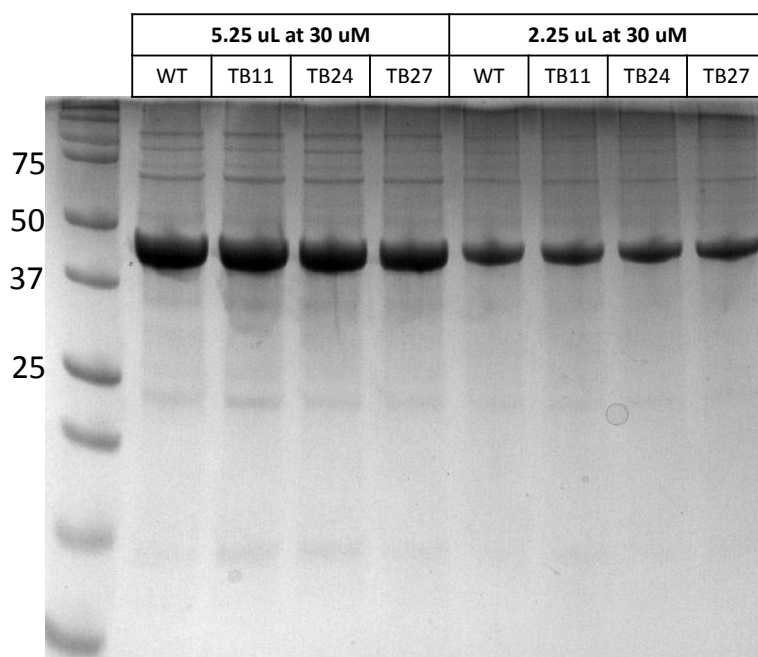


Figure S10. Purified variants stocked at equimolar concentrations.

Original Con1	
DNA sequence	ATGCACCATCACCATCACCATGCCGTAACCACCTTATCAGGTTTATCAGGTG AGCAAGGTCCGTCCGGTGATATGACAACTGAAGAAGATAGTGCTACCCATA TTAAATTCTCAAAACGTGATGAGGACGGCCGTGAGTTAGCTGGTGCAACTA TGGAGTTGCGTGATTCATCTGGTAAAACTATTAGTACATGGATTTCAGATGG ACATGTGAAGGATTTCTACCTGTATCCAGGAAAATATACATTTGTCGAAACC GCAGCACCAGACGGTTATGAGGTAGCAACTCCAATTGAATTTACAGTTAATG AGGACGGTCAGGTTACTGTAGATGGTGAAGCAACTGAAGGTGACGCTCATA CTGGAGGTAGCGGTGGCTCAAGCAAATCTTTATTCCGTGGATTACGCGATT ACAACCCGATCGCCAGCAATATTTGTCATTTGACTAATGAGAGCGACGGTC ATAGTAACTCGTTGTATGGCATCGGGTTTGGGCCGCTTATTATTACAAACCA ACATTTGTTTCGTCGCAACAACGGGGAACCTTACGATTCAATCTCGCCATGG CGAGTTCGTAGTTAAGAACACCACGCAACTTAAGCTTCTGCCAATCGATGG GCGTGATATCCTTATCATTCGTTTGCCCAAAGATTTCACACCTTTCCACAG AAACTTAAATTCCGCCAACCTGAAAAAGGCGAACGCATTTGCTTGGTTGGCT CGAATTTTCAGACCAAGTCCATCACCAGTACAGTCAGCGAAACATCAACAAC CATGCCGGTGGAGAATAGTCAATTTTGAAGCACTGGATTTC AACCAAAGA CGGTCAATTGTGGCCTGCCTCTTGTTCACGAAGGATGGCAAGATCCTGGG CATCATTCTTTGGCGAATTTTACCAATACGATCAACTACTTCGCCGCATT CCAGAAGATTTTGAGGAGACGTATCTTCATACCCAAGAAGCTCAAGAATGG GTAAAGCACTGGAAATATAACCCGGATGCGATCTCCTGGGGGAGCCTTAAC CTTCAAGAAAGCCAACCAGAAGAGCCTTTCAAATCGTAAATAGTAACGG ACGGAGGTAGCGGTGGCTCAGAACAGAACTGATTAGCGAAGAAGATCTGT AA
Protein sequence	MH HHHHHAVTTL SGLSGEQGPSGDMTTEEDSATHIKFSKRDEGRELAGATM ELRDSSGKTISTWISDGHVKDFLYPGKYTFVETAAPDGYEVATPIEFTVNEDG QVTV DGEATEGDAHTGSGSGSSKSLFRGLRDYNPIASNICHLTNESDGHNSL YGIGFGPLIITNQHLFRRNNGELTIQSRHGEFVVKNTTQLKLLPIDGRDILIRLPK DFPPFPQKLKFRQPEKGERICLVGSNFQTKSITSTVSETSTTMPVENSQFWKH WISTKDGHCGPLVSTKD GKILGIHSLANFTNTINYFAAFPEDFEETYLHTQEAQ EWVKHWKYNPD AISWGS LN LQESQPEEPFKIVKL VTDGGSGGSEQKLISEEDL *
Identity	His-SpyCatcher-Con1-myc

Table S1. Annotated Con1 DNA and protein sequence.

FRET substrate	
DNA sequence	ATGCGGGGTTCTCATCATCATCATCATGGTATGGCTAGCATGACTGGT GGACAGCAAATGGGTCCGGATCTGTACGACGATGACGATAAGGATCCGGG CCGCATGGTGAGCAAGGGCGAGGAGCTGTTACCCGGGGTGGTGCCCATC CTGGTCGAGCTGGACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGG CGAGGGCGAGGGCGATGCCACCTACGGCAAGCTGACCCTGAAGTTCATCT GCACCACCGGCAAGCTGCCCCGTGCCCTGGCCCACCCTCGTGACCACCCTG ACCTGGGGCGTGCAAGTCTTCAGCCGCTACCCCGACCACATGAAGCAGCA CGACTTCTTCAAGTCCGCCATGCCCGAAGGCTACGTCCAGGAGCGCACCA TCTTCTTCAAGGACGACGGCAACTACAAGACCCGCGCCGAGGTGAAGTTC GAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAGGGCATCGACTTCAA GGAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACATACATCAGCC ACAACGTCTATATACCGCCGACAAGCAGAAGAACGGCATCAAGGCCAACT TCAAGATCCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCAC TACCAGCAGAACACCCCATCGGCGACGGCCCCGTGCTGCTGCCCCGACAA CCACTACCTGAGCACCCAGTCCGCCCTGAGCAAAGACCCCAACGAGAAGC GCGATCATATGGTCCTGCTGGAGTTCGTGACCGCCGCGGGGATCGGTACC GTAGGATTTCTAACAGCGACCGAAGCAGTGTATCATCAATCCCTGGATTCC

	ATCACCGTTAACGGTACCGGTGGAATGGTGAGCAAGGGCGAGGAGCTGTT CACCGGGGTGGTGCCCATCCTGGTCGAGCTGGACGGCGACGTAAACGGC CACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGATGCCACCTACGGCAA GCTGACCCTGAAGTTCATCTGCACCACCGGCAAGCTGCCCCGTGCCCTGGC CCACCCTCGTGACCACCTTCGGCTACGGCCTGCAGTGCTTCGCCCCGTAC CCCGACCACATGAAGCAGCAGACTTCTTCAAGTCCGCCATGCCCGAAGG CTACGTCCAGGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGAC CCGCGCCGAGGTGAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAG CTGAAGGGCATCGACTTCAAGGAGGACGGCAACATCCTGGGGCACAAGCT GGAGTACAACACTACAACAGCCACAACGTCTATATCATGGCCGACAAGCAGAA GAACGGCATCAAGGTGAACTTCAAGATCCGCCACAACATCGAGGACGGCA GCGTGACGCTCGCCGACCACTACCAGCAGAACACCCCCATCGGCGACGGC CCCGTGCTGCTGCCCCGACAACCACTACCTGAGCTACCAGTCCGCCCTGAG CAAAGACCCCAACGAGAAGCGCGATCACATGGTCCTGCTGGAGTTCGTGA CCGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAAGTAA
Protein sequence	MRGSHHHHHHGMASMTGGQQMGRDLYDDDDKDPGRMVSKGEELFTGVVPI LVELDGDVNGHKFSVSGEGEGDATYGKLT LF ICTTGKLPVPWPTLVTTLTWG VQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRA EV KFEGDTL VNRIELKGIDFKEDGNILGHKLEYN Y ISHNVYITADKQKNGIKANFKIRHNIEDGS VQLADHYQQNTPIGDGPVLLPDNHYLS T QSALS K DPNEKRDHMLLEFVTAAGI GTVGFLTATEAVYHQSLDSITVNGTGGMVSKGEELFTGVVPI L VELDGDVNGH KFSVSGEGEGDATYGKLT LF ICTTGKLPVPWPTLVTTFGYGLQCFARYPDHM KQHDFFKSAMPEGYVQERTIFFKDDGNYKTRA EV KFEGDTLVNRIELKGIDFKE DGNILGHKLEYN Y NSHN V YIMADKQKNGIKVNF K IRHNIEDGSVQLADHYQQNT PIGDGPVLLPDNHYLSYQSALS K DPNEKRDHMLLEFVTAAGITLGMDELYK*
Identity	His-CFP-cut site-YFP

Table S2. Annotated FRET substrate DNA and protein sequence.

Code	Purpose	Sequence
P291	B_1_F	TTAACCATGTGCTGAGCCAACCAGAAGAGCCTT
P292	B_1_R	GGCTCAGCACATGGTTAAGGCTCCCCCAGGA
P293	A_1_F	GAATCGATTGCCAACAACACCAATACGATCAACTACT
P294	A_1_R	GTTGTTGGCAATCGATTTCGATGCCCAGGATCT
P298	B_2_F	TTAACTATCTGATTAGCCAACCAGAAGAGCCTT
P299	B_2_R	GGCTAATCAGATAGTTAAGGCTCCCCCAGGA
P300	B_3_F	TTAACGAAGGCAAAAGCCAACCAGAAGAGCCTT
P301	B_3_R	GGCTTTTGCTTCGTTAAGGCTCCCCCAGGA
P302	B_4_F	TTAACAACATCGCAGCCAACCAGAAGAGCCTT
P303	B_4_R	GGCTGCGATTGTTGTTAAGGCTCCCCCAGGA
P304	B_5_F	TTAACCAAGCCATAGCCAACCAGAAGAGCCTT
P305	B_5_R	GGCTATGGGTCTGGTTAAGGCTCCCCCAGGA
P306	B_6_F	TTAACCGCATGCATAGCCAACCAGAAGAGCCTT
P307	B_6_R	GGCTATGCATGCGTTAAGGCTCCCCCAGGA
P308	B_7_F	TTAACTGGGCGAAAAGCCAACCAGAAGAGCCTT
P309	B_7_R	GGCTTTTGCCCGAGTTAAGGCTCCCCCAGGA
P310	B_8_F	TTAACCATAGCTATAGCCAACCAGAAGAGCCTT
P311	B_8_R	GGCTATAGCTATGGTTAAGGCTCCCCCAGGA
P312	B_9_F	TTAACCATATGTATAGCCAACCAGAAGAGCCTT
P313	B_9_R	GGCTATACATATGGTTAAGGCTCCCCCAGGA
P314	B_10_F	TTAACCTGAACATGAGCCAACCAGAAGAGCCTT
P315	B_10_R	GGCTCATGTTCAAGTTAAGGCTCCCCCAGGA
P317	B_11_R	GGCTGCCCATCGCGTTAAGGCTCCCCCAGGA
P318	B_12_F	TTAACAATTTTGGAGCCAACCAGAAGAGCCTT
P319	B_12_R	GGCTCCAAATTTGTTAAGGCTCCCCCAGGA
P320	B_13_F	TTAACTTTTCATAGCAGCCAACCAGAAGAGCCTT
P321	B_13_R	GGCTGCTATGAAAGTTAAGGCTCCCCCAGGA
P322	B_14_F	TTAACAGCCATTTAGCCAACCAGAAGAGCCTT
P323	B_14_R	GGCTAAAATGGCTGTTAAGGCTCCCCCAGGA
P325	B_15_R	GGCTTCCACCGCGTTAAGGCTCCCCCAGGA
P326	B_16_F	TTAACATGGATCATAGCCAACCAGAAGAGCCTT
P327	B_16_R	GGCTATGATCCATGTTAAGGCTCCCCCAGGA
P328	B_17_F	TTAACATTGTGACCAAGCCAACCAGAAGAGCCTT
P329	B_17_R	GGCTGGTCACAATGTTAAGGCTCCCCCAGGA

P330	B_18_F	TTAACAGCAAACAGAGCCAACCAGAAGAGCCTT
P331	B_18_R	GGCTCTGTTTGCTGTTAAGGCTCCCCAGGA
P332	B_19_F	TTAACGATCATCGCAGCCAACCAGAAGAGCCTT
P333	B_19_R	GGCTGCGATGATCGTTAAGGCTCCCCAGGA
P334	B_20_F	TTAACCGCAGCAAAAGCCAACCAGAAGAGCCTT
P335	B_20_R	GGCTTTTGCTGCGGTTAAGGCTCCCCAGGA
P336	B_21_F	TTAACATTGAAGATAGCCAACCAGAAGAGCCTT
P337	B_21_R	GGCTATCTTCAATGTTAAGGCTCCCCAGGA
P341	B_23_R	GGCTATGGCGCCAGTTAAGGCTCCCCAGGA
P343	B_24_R	GGCTAAAGCGCGGTTAAGGCTCCCCAGGA
P344	B_25_F	TTAACATGGATTATAGCCAACCAGAAGAGCCTT
P345	B_25_R	GGCTATAATCCATGTTAAGGCTCCCCAGGA
P347	B_26_R	GGCTGCTCACGCTGTTAAGGCTCCCCAGGA
P348	B_27_F	TTAACGGCTATTATAGCCAACCAGAAGAGCCTT
P349	B_27_R	GGCTATAATAGCGTTAAGGCTCCCCAGGA
P350	B_28_F	TTAACGGCAACCATAGCCAACCAGAAGAGCCTT
P351	B_28_R	GGCTATGGTTGCCGTTAAGGCTCCCCAGGA
P352	B_29_F	TTAACTTTTCGCTTTAGCCAACCAGAAGAGCCTT
P353	B_29_R	GGCTAAAGCGAAAGTTAAGGCTCCCCAGGA
P354	B_30_F	TTAACCGCACCGCAGCCAACCAGAAGAGCCTT
P355	B_30_R	GGCTCGCGGTGCGGTTAAGGCTCCCCAGGA
P356	B_31_F	TTAACACCAAAAAGAGCCAACCAGAAGAGCCTT
P357	B_31_R	GGCTCTTTTTGGTGTTAAGGCTCCCCAGGA
P358	B_32_F	TTAACTTTAGCCTGAGCCAACCAGAAGAGCCTT
P359	B_32_R	GGCTCAGGCTAAAGTTAAGGCTCCCCAGGA
P360	B_33_F	TTAACCGCAAAAACAGCCAACCAGAAGAGCCTT
P361	B_33_R	GGCTGTTTTTGCGGTTAAGGCTCCCCAGGA
P362	B_34_F	TTAACATGACCTTCAGCCAACCAGAAGAGCCTT
P363	B_34_R	GGCTGAAGGTCATGTTAAGGCTCCCCAGGA
P364	B_35_F	TTAACTATGGCGATAGCCAACCAGAAGAGCCTT
P365	B_35_R	GGCTATCGCCATAGTTAAGGCTCCCCAGGA
P366	B_36_F	TTAACAAAATGATTAGCCAACCAGAAGAGCCTT
P367	B_36_R	GGCTAATCATTTTGTTAAGGCTCCCCAGGA
P368	B_37_F	TTAACGCGAAAACAGCCAACCAGAAGAGCCTT
P369	B_37_R	GGCTGGTTTTTCGCGTTAAGGCTCCCCAGGA
P370	B_38_F	TTAACTTCCTGACCAGCCAACCAGAAGAGCCTT
P371	B_38_R	GGCTGGTCAGGAAGTTAAGGCTCCCCAGGA
P372	B_39_F	TTAACTATGTGATAGCCAACCAGAAGAGCCTT
P373	B_39_R	GGCTATCCACATAGTTAAGGCTCCCCAGGA
P374	B_40_F	TTAACTTCCTGGTGAGCCAACCAGAAGAGCCTT
P375	B_40_R	GGCTCACCAGGAAGTTAAGGCTCCCCAGGA
P376	B_41_F	TTAACATTGAAAACAGCCAACCAGAAGAGCCTT
P377	B_41_R	GGCTGTTTTCAATGTTAAGGCTCCCCAGGA
P378	B_42_F	TTAACGTGGCGAACAGCCAACCAGAAGAGCCTT
P379	B_42_R	GGCTGTTCCGCACGTTAAGGCTCCCCAGGA
P380	B_43_F	TTAACATGCAGCATAGCCAACCAGAAGAGCCTT
P381	B_43_R	GGCTATGCTGCATGTTAAGGCTCCCCAGGA
P382	B_44_F	TTAACATTAGCGATAGCCAACCAGAAGAGCCTT
P383	B_44_R	GGCTATCGCTAATGTTAAGGCTCCCCAGGA
P384	B_45_F	TTAACGAAGTGACCAGCCAACCAGAAGAGCCTT
P385	B_45_R	GGCTGGTCACTTCGTTAAGGCTCCCCAGGA
P386	B_46_F	TTAACAACCAGGTGAGCCAACCAGAAGAGCCTT
P387	B_46_R	GGCTCACCTGGTTGTTAAGGCTCCCCAGGA
P388	B_47_F	TTAACAGCGATACCAGCCAACCAGAAGAGCCTT
P389	B_47_R	GGCTGGTATCGCTGTTAAGGCTCCCCAGGA
P390	B_48_F	TTAACATGCGCCTGAGCCAACCAGAAGAGCCTT
P391	B_48_R	GGCTCAGGCGCATGTTAAGGCTCCCCAGGA
P392	A_2_F	ACCTCGCATGCCAACACCACCAATACGATCAACTACT
P393	A_2_R	GGTGTTGGCATGCGAGGTGATGCCCAGGATCT
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P395	A_3_R	AAAGTTGGCCATCGAACGGATGCCCAGGATCT
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P398	A_5_F	CGCTCGCGCGCCAACCTGACCAATACGATCAACTACT
P399	A_5_R	CAGGTTGGCGCGCGAGCGGATGCCCAGGATCT
P400	A_6_F	TGGTCGGCGGCCAACAAAACCAATACGATCAACTACT
P401	A_6_R	TTTGTTGGCCGCCGACCAGATGCCCAGGATCT
P402	A_7_F	CGCTCGAAAGCCAACCTGACCAATACGATCAACTACT
P403	A_7_R	CAGGTTGGCTTTCGAGCGGATGCCCAGGATCT
P404	A_8_F	TGGTCGGCGGCCAACTTTACCAATACGATCAACTACT

P405	A 8 R	AAAGTTGGCCGCCGACCAGATGCCCAGGATCT
P406	A 9 F	GGCTCGTTTGCCAACTTTACCAATACGATCAACTACT
P407	A 9 R	AAAGTTGGCAAACGAGCCGATGCCCAGGATCT
P408	A 10 F	CAGTCGCTGGCCAACTTTACCAATACGATCAACTACT
P409	A 10 R	AAAGTTGGCCAGCGACTGGATGCCCAGGATCT
P410	A 11 F	CATTGACCGCCAACGTGACCAATACGATCAACTACT
P411	A 11 R	CACGTTGGCGGTGGAATGGATGCCCAGGATCT
P412	A 12 F	GGCTCGATGCCAAACGCCAATACGATCAACTACT
P413	A 12 R	GCGGTTGGCATAACGAGCCGATGCCCAGGATCT
P414	A 13 F	GTGTCGATGGCCAACAACACCAATACGATCAACTACT
P415	A 13 R	GTTGTTGGCCATCGACACGATGCCCAGGATCT
P416	A 14 F	GTGTCGACCGCCAACGATACCAATACGATCAACTACT
P417	A 14 R	ATCGTTGGCGGTGACACGATGCCCAGGATCT
P418	A 15 F	TATTCGTTTGCCAAACAACCAATACGATCAACTACT
P419	A 15 R	GTTGTTGGCAAACGAATAGATGCCCAGGATCT
P420	A 16 F	TGGTCGCTGGCCAACGGCACCAATACGATCAACTACT
P421	A 16 R	GCCGTTGGCCAGCGACCAGATGCCCAGGATCT
P422	A 17 F	CAGTCGCGGCCAACGAAACCAATACGATCAACTACT
P423	A 17 R	TTCGTTGGCGCGCGACTGGATGCCCAGGATCT
P424	A 18 F	GAATCGCGCGCCAACTGGACCAATACGATCAACTACT
P425	A 18 R	CCAGTTGGCGCGGATTTCGATGCCCAGGATCT
P426	A 19 F	ATGTCGAGCGCCAACATTACCAATACGATCAACTACT
P427	A 19 R	AATGTTGGCGCTCGACATGATGCCCAGGATCT
P428	A 20 F	GATTCGCTGGCCAACGGCACCAATACGATCAACTACT
P429	A 20 R	GCCGTTGGCCAGCGAATCGATGCCCAGGATCT
P430	A 21 F	GAATCGGATGCCAACTGGACCAATACGATCAACTACT
P431	A 21 R	CCAGTTGGCATCCGATTTCGATGCCCAGGATCT
P432	A 22 F	AAATCGAGCGCCAACCTTTACCAATACGATCAACTACT
P433	A 22 R	AAAGTTGGCGCTCGATTTCGATGCCCAGGATCT
P434	A 23 F	AAATCGATTGCCAACTGGACCAATACGATCAACTACT
P435	A 23 R	CCAGTTGGCAATCGATTTCGATGCCCAGGATCT
P438	B 11 F 2	TTAACGCGATGGGCAGCCAACCAAGAGGCC
P439	B 15 F 2	TTAACGCGGTGGAAGCCAACCAAGAGGCC
P440	B 22 F 2	TTAACGCGCTGTGAGCCAACCAAGAGGCC
P441	B 22 R 2	GGCTCACACGTGCGTTAAGGCTCCCCAGGA
P442	B 23 F 2	TTAACTGGCGCCATAGCCAACCAAGAGGCC
P443	B 24 F 2	TTAACGCGCGCTTTAGCCAACCAAGAGGCC
P444	B 26 F 2	TTAACAGCGTGAGCAGCCAACCAAGAGGCC
P445	A 24 F 2	GACAGCGCAGCCAACGGCACCAATACGATCAACTACT
P447	A 24 R 2	GCCGTTGGCTGCGCTGTGATGCCCAGGATCT
P448	A 25 F	GGCTCGAGCGCCAACATACCAATACGATCAACTACT
P449	A 25 R	ATAGTTGGCGCTCGAGCCGATGCCCAGGATCT
P450	A 26 F	CTGTCGACCGCCAACAAAACCAATACGATCAACTACT
P451	A 26 R	TTTGTTGGCGGTGACAGGATGCCCAGGATCT
P452	A 27 F	CGTTGCGGCGCCAACCATACCAATACGATCAACTACT
P453	A 27 R	ATGGTTGGCGCGCGAACGGATGCCCAGGATCT
P454	A 28 F	CATTGATGGCCAACGAAACCAATACGATCAACTACT
P455	A 28 R	TTCGTTGGCCATCGAATGGATGCCCAGGATCT
P456	A 29 F	CAGTCGGTGGCCAACACCACCAATACGATCAACTACT
P457	A 29 R	GGTGTGGCCACCGACTGGATGCCCAGGATCT
P458	A 30 F	AACTCGAAAGCCAACGATACCAATACGATCAACTACT
P459	A 30 R	ATCGTTGGCTTTGAGTTGATGCCCAGGATCT
P460	A 31 F	GAATCGCATGCCAACAGCACCAATACGATCAACTACT
P461	A 31 R	GCTGTTGGCATGCGATTTCGATGCCCAGGATCT
P462	A 32 F	AGCTCGCGTGCCAACCTTTACCAATACGATCAACTACT
P463	A 32 R	AAAGTTGGCACGCGAGCTGATGCCCAGGATCT
P464	A 33 F	TGGTCGCGCGCCAACGAAACCAATACGATCAACTACT
P465	A 33 R	TTCGTTGGCGCGCGACCAGATGCCCAGGATCT
P466	A 34 F	GCGTCGCTGGCCAACCTTTACCAATACGATCAACTACT
P467	A 34 R	AAAGTTGGCCAGCGACGCGATGCCCAGGATCT
P468	A 35 F	AGCTCGTGGCCAACAGCACCAATACGATCAACTACT
P469	A 35 R	GCTGTTGGCCACGAGCTGATGCCCAGGATCT
P470	A 36 F	ATTTGGAAGCCAACAACCAATACGATCAACTACT
P471	A 36 R	GTTGTTGGCTTCCGAAATGATGCCCAGGATCT
P472	A 37 F	ATTTGCTGGCCAACCATACCAATACGATCAACTACT
P473	A 37 R	ATGGTTGGCCAGCGAAATGATGCCCAGGATCT
P474	A 38 F	CTGTCGATGGCCAACGATACCAATACGATCAACTACT
P475	A 38 R	ATCGTTGGCCATCGACAGGATGCCCAGGATCT
P476	A 39 F	ATTTGCGCGGCCAACACCAATACGATCAACTACT
P477	A 39 R	GTTGTTGGCCGCCGAAATGATGCCCAGGATCT

P478	A_40_F	CTGTCGATGGCCAACTTTACCAATACGATCAACTACT
P479	A_40_R	AAAGTTGGCCATCGACAGGATGCCCAGGATCT
P480	A_41_F	ATTTCGTATGCCAACATTACCAATACGATCAACTACT
P481	A_41_R	AATGTTGGCATACGAAATGATGCCCAGGATCT
P482	A_42_F	AGCTCGACCGCCAACGAAACCAATACGATCAACTACT
P483	A_42_R	TTCGTTGGCGGTGCGAGCTGATGCCCAGGATCT
P484	A_43_F	GATTCGGATGCCAACACCACCAATACGATCAACTACT
P485	A_43_R	GGTGTGGCATCCGAATCGATGCCCAGGATCT
P486	A_44_F	GAATCGCAGGCCAACAAAACCAATACGATCAACTACT
P487	A_44_R	TTTGTGGCCTGCGATTTCGATGCCCAGGATCT
P488	A_45_F	AACTCGCAGGCCAACTATACCAATACGATCAACTACT
P489	A_45_R	ATAGTTGGCCTGCGAGTTGATGCCCAGGATCT
P490	A_46_F	AGCTCGGATGCCAACCGCACCAATACGATCAACTACT
P491	A_46_R	GCGGTTGGCATCCGAGCTGATGCCCAGGATCT
P492	A_47_F	CAGTCGCTGGCCAACGGCACCAATACGATCAACTACT
P493	A_47_R	GCCGTTGGCCAGCGACTGGATGCCCAGGATCT
P494	A_48_F	AAATCGGATGCCAACATTACCAATACGATCAACTACT
P495	A_48_R	AATGTTGGCATCCGATTGATGCCCAGGATCT
P501	TB_1_F	TTAACGTGAGCGACAGCCAACCAGAAGAGCCTT
P502	TB_1_R	GGCTGTCGCTCACGTTAAGGCTCCCCAGGA
P503	TB_2_F	TTAACTTCAGCTATAGCCAACCAGAAGAGCCTT
P504	TB_2_R	GGCTATAGCTGAAGTTAAGGCTCCCCAGGA
P505	TB_3_F	TTAACTTCAGCACCAGCCAACCAGAAGAGCCTT
P506	TB_3_R	GGCTGGTGCTGAAGTTAAGGCTCCCCAGGA
P507	TB_4_F	TTAACTTCACCACGAGCCAACCAGAAGAGCCTT
P508	TB_4_R	GGCTCGTGGTGAAGTTAAGGCTCCCCAGGA
P509	TB_5_F	TTAACGTGAGCTACAGCCAACCAGAAGAGCCTT
P510	TB_5_R	GGCTGTAGCTCACGTTAAGGCTCCCCAGGA
P511	TB_6_F	TTAACTTCAGGAAAGCCAACCAGAAGAGCCTT
P512	TB_6_R	GGCTTTCCTGGAAGTTAAGGCTCCCCAGGA
P513	TB_7_F	TTAACTTTACCTATAGCCAACCAGAAGAGCCTT
P514	TB_7_R	GGCTATAGGTAAAGTTAAGGCTCCCCAGGA
P515	TB_8_F	TTAACATGAGCTATAGCCAACCAGAAGAGCCTT
P516	TB_8_R	GGCTATAGCTCATGTTAAGGCTCCCCAGGA
P517	TB_9_F	TTAACTTCGCATATAGCCAACCAGAAGAGCCTT
P518	TB_9_R	GGCTATATGCGAAGTTAAGGCTCCCCAGGA
P519	TB_10_F	TTAACGTTAGCACCAGCCAACCAGAAGAGCCTT
P520	TB_10_R	GGCTGGTGCTAACGTTAAGGCTCCCCAGGA
P521	TB_11_F	TTAACTTCAGCCCAGCCAACCAGAAGAGCCTT
P522	TB_11_R	GGCTCGGGCTGAAGTTAAGGCTCCCCAGGA
P523	TB_12_F	TTAACGTGCAGACCAGCCAACCAGAAGAGCCTT
P524	TB_12_R	GGCTGGTCTGCACGTTAAGGCTCCCCAGGA
P525	TB_13_F	TTAACATGCAGTATAGCCAACCAGAAGAGCCTT
P526	TB_13_R	GGCTATACTGCATGTTAAGGCTCCCCAGGA
P527	TB_14_F	TTAACTTTAGCGAAAGCCAACCAGAAGAGCCTT
P528	TB_14_R	GGCTTTCGCTAAAGTTAAGGCTCCCCAGGA
P529	TB_15_F	TTAACTTTGTGAAAGCCAACCAGAAGAGCCTT
P530	TB_15_R	GGCTTTCACAAAGTTAAGGCTCCCCAGGA
P531	TB_16_F	TTAACTTTGCAACCAGCCAACCAGAAGAGCCTT
P532	TB_16_R	GGCTGGTTGCAAAGTTAAGGCTCCCCAGGA
P533	TB_17_F	TTAACGTGAGCGAAAGCCAACCAGAAGAGCCTT
P534	TB_17_R	GGCTTTCGCTCACGTTAAGGCTCCCCAGGA
P535	TB_18_F	TTAACATGCAGGAAAGCCAACCAGAAGAGCCTT
P536	TB_18_R	GGCTTTCCTGCATGTTAAGGCTCCCCAGGA
P537	TB_19_F	TTAACATGAGCACGAGCCAACCAGAAGAGCCTT
P538	TB_19_R	GGCTCGTGCTCATGTTAAGGCTCCCCAGGA
P539	TB_20_F	TTAACTTCGCAGAAAGCCAACCAGAAGAGCCTT
P540	TB_20_R	GGCTTTCGCGAAGTTAAGGCTCCCCAGGA
P541	TB_21_F	TTAACTTCCAGTATAGCCAACCAGAAGAGCCTT
P542	TB_21_R	GGCTATACTGGAAGTTAAGGCTCCCCAGGA
P543	TB_22_F	TTAACGTGAGCTTTAGCCAACCAGAAGAGCCTT
P544	TB_22_R	GGCTAAAGCTCACGTTAAGGCTCCCCAGGA
P545	TB_23_F	TTAACATGAGCGAAAGCCAACCAGAAGAGCCTT
P546	TB_23_R	GGCTTTCGCTCATGTTAAGGCTCCCCAGGA
P547	TB_24_F	TTAACTTTAGCGATAGCCAACCAGAAGAGCCTT
P548	TB_24_R	GGCTATCGCTAAAGTTAAGGCTCCCCAGGA
P549	TB_25_F	TTAACGTGCAGTATAGCCAACCAGAAGAGCCTT
P550	TB_25_R	GGCTATACTGCACGTTAAGGCTCCCCAGGA
P551	TB_26_F	TTAACTTTAGCCATAGCCAACCAGAAGAGCCTT
P552	TB_26_R	GGCTATGGCTAAAGTTAAGGCTCCCCAGGA

P553	TB_27_F	TTAACTTTAGCAACAGCCAACCAGAAGAGCCTT
P554	TB_27_R	GGCTGTTGCTAAAGTTAAGGCTCCCCCAGGA
P555	TB_28_F	TTAACTTTAGCGGCAGCCAACCAGAAGAGCCTT
P556	TB_28_R	GGCTGCCGCTAAAGTTAAGGCTCCCCCAGGA
P557	TB_29_F	TTAACTTCAGCTTTAGCCAACCAGAAGAGCCTT
P558	TB_29_R	GGCTAAAGCTGAAGTTAAGGCTCCCCCAGGA
P559	TB_30_F	TTAACATGAGCCCGAGCCAACCAGAAGAGCCTT
P560	TB_30_R	GGCTCGGGCTCATGTTAAGGCTCCCCCAGGA
P561	TB_31_F	TTAACTTTCTGGAAAGCCAACCAGAAGAGCCTT
P562	TB_31_R	GGCTTTCCAGAAAGTTAAGGCTCCCCCAGGA
P563	TB_32_F	TTAACATGGTGGAAGCCAACCAGAAGAGCCTT
P564	TB_32_R	GGCTTTCCACCATGTTAAGGCTCCCCCAGGA
P565	TA_1_F	CATTGCTGGCCAACCAGACCAATACGATCAACTACT
P566	TA_1_R	CTGGTTGGCCAGCGAATGGATGCCCAGGATCT
P567	TA_2_F	CATTGCTGGCCAACCCGACCAATACGATCAACTACT
P568	TA_2_R	CGGGTTGGCCAGCGAATGGATGCCCAGGATCT
P569	TA_3_F	CATTGCTGGCCAACGACCAATACGATCAACTACT
P570	TA_3_R	GCAGTTGGCCAGCGAATGGATGCCCAGGATCT
P571	TA_4_F	CATTGCGCGCCAACTTCACCAATACGATCAACTACT
P572	TA_4_R	GAAGTTGGCCGGCGAATGGATGCCCAGGATCT
P573	TA_5_F	CATTGCTGGCCAACGATACCAATACGATCAACTACT
P574	TA_5_R	ATCGTTGGCCAGCGAATGGATGCCCAGGATCT
P575	TA_6_F	CAGTCGCTGGCCAACGTGACCAATACGATCAACTACT
P576	TA_6_R	CACGTTGGCCAGCGACTGGATGCCCAGGATCT
P577	TA_7_F	CATTGCTGGCCAACGACCAATACGATCAACTACT
P578	TA_7_R	CCAGTTGGCCAGCGAATGGATGCCCAGGATCT
P579	TA_8_F	CATTGCTGGCCAACCTGACCAATACGATCAACTACT
P580	TA_8_R	CAGGTTGGCCAGCGAATGGATGCCCAGGATCT
P581	TA_9_F	CATTGCTGGCCAACATACCAATACGATCAACTACT
P582	TA_9_R	ATAGTTGGCCAGCGAATGGATGCCCAGGATCT
P583	TA_10_F	CATTGCTGGCCAACAAAACCAATACGATCAACTACT
P584	TA_10_R	TTTGTTGGCCAGCGAATGGATGCCCAGGATCT

Table S3. Oligonucleotides used in this study.